Audit, step 1: brainstorm of everything that could go wrong

Author: Claude (the same model that wrote the code; that's why step 2 is an independent review). **Date:** 2026-10-20 (night). **Scope:** Knotted Coil Designer v1.7: geometry, quasi-static EM, solvers, full-wave co-simulation, app, validation, documentation, integrity, packaging.

Legend. *Covered* = an existing check would catch it. *Partial* = some protection. *Open* = nothing would catch it today.

A. Physics and modelling assumptions

#	Failure mode	Consequence	Detection	Status
A1	Quasi-static coupling k ignores capacitors, tissue load, wave effects	Decoupling nulls in the wrong place on a real coil	V19 (full-wave pair), bench S21	Partial (V19 tonight; no bench)
A2	Uniform-current B1 assumption	B1 maps wrong, especially at 7T and at depth	V18 (Tier 2)	Partial
A3	Self-inductance formula assumes skin-effect surface current, $a \ll R$	L error for thick wire or tight bends	V2 (circle only)	Partial: no check on tight knot bends
A4	Crossing lift modelled as a sinusoid in z, while real wire lifts only locally	Small L and M error; clearance check may pass for a shape that cannot be wound	Clearance check	Open: fabrication realism not validated
A5	Kernel softening ε between elements	Biases M for heavily overlapping elements	none	Open: sensitivity untested (logged)
A6	Wrapping a flat knot onto the helmet distorts it	Topology could change on high-curvature helmets; distortion only reported as a length change	Wrap distortion %	Open: crossings not re-checked after wrapping
A7	FDTD staircasing; thin wire has an effective radius ≠ real radius	Inductance and resonance off (**seen: V15 −9.5%**)	V15, mesh sensitivity	Being investigated
A8	Lumped port along one mesh axis, while the wire tangent is oblique	Gap geometry and parasitic capacitance wrong	none	Open
A9	Phantom tissue values: 7T secondary source; 3T an approximation from body-phantom values	Loaded Q, capacitor values and B1 at depth off	Labelled	Open: unverified values
A10	Phantom geometry (ellipsoid, layer thicknesses) is a placeholder	Loading and B1 depth off	none	Open
A11	Preamp modelled as a fixed 2 Ω; matching topology assumed	Decoupling estimates differ from real preamps	none	Open (placeholder)
A12	Capacitor ESR and inductor Q placeholders	Q and noise estimates off	none	Open
A13	Copper σ, skin depth; no proximity effect at crossings	Loss underestimated	none	Open
A14	B0 assumed along z; $B1^-$ uses the (Hx − iHy) convention	$B1^+/B1^-$ swapped if the sign convention differs from openEMS's	Same convention as openEMS tutorial	Partial
A15	The voxel phantom and our analytic head use different "inside" tests	V18 near-surface points misclassified	none	Open

#	Failure mode	Consequence	Detection	Status
A16	Coverage and regularity metrics are proxies for image quality (SNR, g-factor)	Optimising the wrong objective	none	Open (by design; disclosed)

B. Numerics

#	Failure mode	Consequence	Detection	Status
B1	Neumann near-field refinement insufficient when wires cross between elements	M error for overlapping elements	V4 (one case)	Partial
B2	Self-L with too few points per wire radius on tight bends	L error	V2 at 400 points (circle)	Partial
B3	Adaptive point count capped at 102,400; stable-count rule fooled again	Spurious crossings → wrong topology verdict	V7 history	Partial
B4	Crossing detection: segments parallel or overlapping (den ≈ 0) skipped	Missed crossings in degenerate cases	none	Open
B5	Lift bisection assumes clearance is monotone in h	Wrong h, or “infeasible” reported wrongly	none	Open
B6	Elliptic-integral AGM precision near k → 1	V1 error at small separations	V1 (d ≥ 5 mm)	Partial
B7	Stage-1 table interpolation error; table eps tied to the initial Rf	Wrong optimum in fast mode	Stage 2, final exact report	Partial
B8	Co-sim root search: multiple roots; picking “nearest first-pass” may choose a non-physical branch	Wrong capacitor values	V17 (S11 only)	Partial: a matched but wrong branch would pass
B9	FDTD end criterion and ring-down; port de-embedding	S errors near resonance	V16 (passivity/reciprocity)	Partial
B10	Frequency interpolation of S (linear) between FDTD samples	Small errors	none	Open
B11	Field superposition normalisation (uf_inc at f0 interpolated)	B1 amplitude and phase errors between ports	none	Open: could test that superposed fields reproduce a direct run
B12	Exact-segment B1 formula singular on the line extension (den → 0 skipped)	Missing contributions at special points	V14	Covered for test points

C. Code correctness (bugs)

#	Failure mode	Consequence	Detection	Status
C1	Unit mix-ups (mm/m, nH/H, pF/F, µT/T), especially at the Python ↔ JS boundary	Wrong values everywhere	V8, V11, V14, V15	Partial
C2	Sign and direction conventions (current direction, mirror reversal, port direction)	Wrong coupling sign or superposition	V9	Partial (geometry only)
C3	Ring layout index bookkeeping (twins, midline)	Asymmetric arrays	V9	Covered
C4	Multi-start selection compares runs with different definitions	Wrong "best"	none	Open
C5	Free-mode zone penalty uses a 1.1·Rf disc; wires reaching beyond it	Zone violations	V12	Covered (final report counts real wires)
C6	App: stale worker results overwriting newer ones; races between element, pair, solve and B1 workers	Wrong numbers displayed	none	Open: no UI tests
C7	App: loading a JSON from an older version (missing fields)	Crashes or silent defaults	none	Open
C8	Exports: port CSV mapping flat → wrapped index assumes identical sampling	Ports at wrong positions	none	Open: could cross-check against the exported case
C9	export_case.js wraps at 600 points while the solver uses 300	Inconsistent geometry between quasi-static and full-wave	none	Open
C10	cut_wire: breaks near the polyline seam, or two breaks closer than the gap	Malformed wires or overlapping ports	none	Open
C11	Lumped port terminals not on mesh lines (edges2grid)	Port geometry altered silently	none	Open
C12	`reduce()` ill-conditioned (huge "open" termination 1e9 Ω)	Numerical noise	none	Open
C13	Hidden global state in KC modules shared across workers	Inconsistent defaults	none	Partial
C14	`validation_status.js` and the reports go stale relative to the code (partial `KC_ONLY` runs overwrite the full report)	App shows wrong validation status	git hash in report	**Open: happened** (a partial run was committed in 5f60ace)

D. Validation design (are the checks themselves right?)

#	Failure mode	Consequence	Status
D1	Checks testing implementation against itself (circular)	False confidence	Partial: V14 fixed this for B1; V15 calibration risk
D2	Thresholds JUDG, not justified (0.5%, 20%, −15 dB…)	Pass/fail arbitrary	Open (disclosed)
D3	Harness changes after seeing results	Moving goalposts	Logged each time; criteria unchanged; still a risk
D4	Test cases unrepresentative (circles, single trefoil) of arrays actually used	Bugs in other regimes hidden	Open
D5	Tuning parameters on C2 configurations (development overlap)	Biased comparison	Partially avoided, disclosed
D6	V3 permanently NOT EVALUABLE	The classic overlap reference is never checked	Open (needs the paper)
D7	No bench validation at all	Everything stays exploratory	Open
D8	Pre-registration commit order not verifiable by readers	Integrity claim weaker	Git history present

E. Research integrity, provenance, documentation

#	Failure mode	Status
E1	Unverified values creeping in as “facts” (σ_Cu, tissue values, λ/10, phantom layers)	Labelled; σ_Cu CONV without a cited source
E2	Documents disagree with code (e.g. Design §10 architecture outdated; README validation table stale)	**Open: §10 is outdated**
E3	AI-authorship headers missing on new files	To check
E4	Third-party licences (openEMS GPL, mariepy GPL, Magpylib BSD, Three.js MIT) not listed	Partial: README lists Three.js only
E5	Personal data in the package (email in git)	Disclosed
E6	IP: sharing before an OIP disclosure	Disclosed; Reza’s decision
E7	Decisions taken without Reza (field strength, matching, installs)	Logged; need confirmation

F. Environment, reproducibility, packaging

#	Failure mode	Status
F1	Solver paths hard-coded to this machine	Overridable by environment variable; documented?
F2	Library versions not pinned in a requirements file	Open
F3	openEMS Windows performance warning (their docs mention MSVC slowdowns)	Noted
F4	Large simulation outputs in OneDrive	Work files kept outside; outputs in `sim/out` could grow
F5	`file://` launch untested	Open
F6	Browser memory or time with large N (wires, tubes, STL)	Open
F7	Python 3.14-only openEMS wheels	Documented?
F8	mariepy pulls in a heavy dependency (PyTorch); immature	Noted

G. Human factors and misuse

#	Failure mode	Status
G1	Users read exploratory numbers (capacitor values, k) as build-ready	Labels present; the fabrication sheet must repeat them
G2	Placeholders forgotten (wire, helmet, zones, tissue)	Shown in UI headings; easy to miss
G3	“Decoupled” verdict misunderstood (which definition?)	Three verdicts shown; could confuse